POMDP-based Object Search with Growing State Space and Hybrid Action Domain

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Research Questions

- How can robots efficiently search for target objects in cluttered 3D environments with growing state spaces and hybrid action domains?
- How can belief tree reuse and neural process filtering improve POMDP-based object search performance?
- How can k-center clustering discretization handle high-dimensional continuous action spaces in POMDP solvers?

Core Idea

This paper tackles robot object search in cluttered environments by formulating it as a POMDP with continuously growing state space (as new objects are detected) and hybrid continuous-discrete actions. The key insight is to combine three techniques: (1) neural process networks to filter out useless primitive actions from the continuous domain, (2) k-center clustering to hierarchically discretize the remaining action space into hyperspheres, and (3) belief tree reuse to avoid rebuilding the entire search tree when new objects appear. A "guessed target object" with grid-world representation provides initial exploration guidance when limited information is available.

Key Formula

Modified UCB for MCTS

$$U(node_o, a) = \hat{Q}(node_o, a) + \omega_1 \sqrt{\frac{\log N(node_o)}{N(node_o, a)}} + \omega_2 range_i + f_{option}(\omega_3 BD)$$

Modified upper confidence bound for action selection where \hat{Q} is average reward, N are visit counts, range_i is hypersphere radius, and BD is belief distance term for growing state spaces. The ω coefficients balance exploration vs exploitation with geometric constraints.

Grid odds update

$$\log Odd(g_j | z_{1:t}) = \log Odd(g_j | z_t) + \log Odd(g_j | z_{1:t-1})$$

Log-odds update for grid cells g_j of the guessed target object based on observations z_{1:t}. This maintains probabilistic beliefs about object locations in a grid world representation, similar to occupancy mapping but for object detection.

Key Figure

techniques like Voronoi optimistic optimization (Lim et al., 2021) or Bayesian optimization (Mern et al., 2021). These methods commonly apply the UCB1 (Auer et al., 2002) algorithm for exploration and Monte Carlo backups for value estimation, differing mainly in how they refine the action domain. Despite advances, the challenges posed by growing state spaces and continuous state, action, and observation spaces remain formidable.

2.4 Neural process in robotics

As a more efficient alternative to Gaussian processes, neural processes excel in function regression by capturing uncertainty in predictions as a stochastic process. In numerous robotics tasks, the ability to account for uncertainty in predictions is crucial for enhancing system robustness and practicality. For instance, (Chen et al., 2022) introduces a conditional neural process (CNP)-based meta-learning algorithm to estimate grasp points from limited depth image trials. The versatility of CNPs is further demonstrated in applications such as 6D pose estimation (Li et al., 2022) and understanding social aspects of navigation (Yildirim et al., 2022). In our work, we utilize neural processes to address uncertain scoring function regression and primitive action filtering, creating a robust framework for decision-making in complex, dynamic environments. This idea was publicly released in our earlier work (Chen et al., 2024b), preceding several concurrent studies exploring neural process-based planning (Jain et

These measurements are processed into state-space information through steps like robot pose estimation, point cloud filtering and segmentation, principal component analysis, object matching, and move-ability estimation. In the filtering stage, a particle filter updates the belief and prunes the belief tree using processed sensor data. The guessed target object and its grid world are updated via log-odds, similar to occupancy grid mapping. Robot pose estimation uses iterative closest point (ICP), initialized by adaptive Monte Carlo localization (AMCL). If the terminal condition is met, the task ends with the cumulative reward; otherwise, the updated belief reuses the existing belief tree and resamples particles to proceed with the MCTS. These steps, as shown in Fig. 2, collectively achieve object search and retrieval.

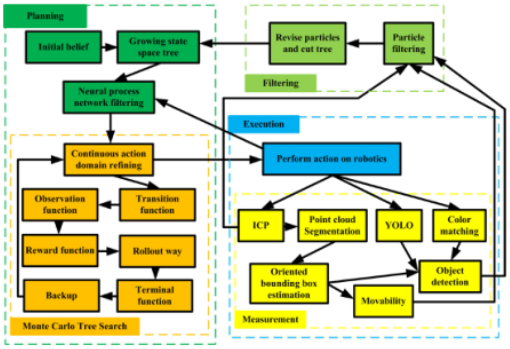

Figure 2. The main steps in our proposed approach, GNPF-kCT

Figure 2: Shows the complete GNPF-kCT framework pipeline with planning, execution, observation, and filtering stages, illustrating how neural process filtering, k-center clustering, and belief tree reuse work together.

Limitations

- Assumes 100% action success rate which is unrealistic even in simulation environments
- Neural process filtering requires domain-specific training data and may not generalize across different environments
- Belief tree reuse approximation may degrade performance when belief differences between old and new states are large
- Move-ability estimation is not very efficient and relies on simple heuristics rather than learned models